



# **Building America's AI Future: A Comprehensive Action Plan for Innovation, Security, and Workforce Development**

## **University at Albany's Response to OSTP's RFI on the Development of an Artificial Intelligence (AI) Action Plan**

### **Executive Summary**

The University at Albany appreciates the opportunity to respond to the Office of Science and Technology Policy (OSTP) request for input related to the Development of an Artificial Intelligence (AI) Action Plan. The dramatic recent impact of AI is the product of multiple synergistic scientific advances that have been facilitated by the U.S. Government over the past decades. We argue that AI is not a single technology, but rather an ecosystem of interconnected technologies and application domains comprising tightly-coupled components such as (i) scientific advances related to the next-generation AI algorithms; (ii) the cutting-edge technologies for the computing, data storage, and computer communication infrastructures needed to support these powerful AI algorithms; (iii) AI deployment in various domains: healthcare, manufacturing, managing critical infrastructures (such as power grids and telecommunications), government operations, and education. We strongly advocate that any policies adopted at the federal level recognize and support all components that inform this AI ecosystem.

Proceeding from this understanding, we have articulated an AI action plan with the following major policy objectives: (a) Clear and straightforward policies that would not inhibit but rather promote faster and better scientific innovations in the core areas of the AI ecosystem: AI algorithms and the associated infrastructures (computing, data storage, and communication) needed for AI; (b) Domain-specific guidelines, instead of generalized regulations, that would encourage faster adoption and incorporation of AI ecosystem; (c) "No Harm for Americans Due to AI" policies to promote wide-spread acceptance of AI and ensure the commercial stability of AI investments; (d) Multi-disciplinary workforce development spanning all the components in the AI ecosystem so that America can sustain and extend its leadership in AI innovation. For achieving these policy objectives, we strongly advocate a coordinated action plan to galvanize strong cooperation across the broad landscape of the AI ecosystem: industries, government agencies, and educational institutions.

## AI at UAlbany: A National Leader in AI Innovation

Launched in 2022 with a \$75 million investment from New York State, AI Plus is a university-wide initiative integrating artificial intelligence across UAlbany's research and academic enterprise, from data science and semiconductor design to philosophy and the arts. The AI Plus Institute, home to over 130 faculty experts, has led UAlbany's largest-ever faculty expansion, solidifying its position as a national leader in AI. In Fall 2024, the university unveiled the most advanced AI supercomputer in the SUNY system, powered by 24 NVIDIA DGX systems and 192 NVIDIA A100 GPUs, making it one of the most capable AI clusters in U.S. higher education. UAlbany is also the first university globally to access IBM's AIU System and the first higher-ed institution to use NVIDIA DGX AI Cloud, demonstrating leadership in next-generation AI infrastructure.

UAlbany drives national AI and semiconductor advancements through strategic partnerships and federally backed initiatives. As a key partner in the \$825 million National Semiconductor Technology Center (NSTC), co-located with the Albany NanoTech Complex—the nation's largest and most advanced non-profit semiconductor R&D facility operated by NY CREATES—UAlbany's College of Nanotechnology, Science, and Engineering (CNSE) is at the forefront of U.S. semiconductor and AI research and workforce development. The university is also a founding member of the \$285 million CHIPS Manufacturing USA Institute for Digital Twins and the \$40 million NORDTECH defense technology consortium, ensuring AI and semiconductor innovations bolster national security and economic growth.

Further solidifying its leadership in AI research and policy, UAlbany is a founding member of the NIST-led U.S. Artificial Intelligence Safety Institute Consortium (AISIC), a federal initiative aimed at advancing AI safety and governance. With a steadfast commitment to AI-driven research, workforce development, and U.S. global competitiveness, UAlbany is a critical partner in shaping the future of AI policy and innovation.

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## Introduction

Artificial Intelligence holds the potential to precipitate the most profound transformation of the global labor market since the Industrial Revolution. While the specific needs generated by the AI revolution we are witnessing cannot be determined with decisive clarity, we believe that an effective national AI policy dedicated to ensuring U.S. competitiveness should not only anticipate the pending disruptions to the American workforce but also demonstrate a flexibility of design that responds to such challenges with an ambitious and structured agility.

## The Plurality of Artificial Intelligence



**Figure 1. AI Ecosystem and Its Plurality**

We argue that AI is not a single technology, but rather an ecosystem of interconnected technological advancements across multiple areas. These include algorithms that power deep learning models, computing infrastructure, high-speed communication technologies, data storage, and memory architectures. Together, these components frame a transformative architecture that enables AI to be deployed across myriad fields (Figure 1). One significant element of this multifaceted understanding of AI lies in the fact that a disruptive innovation in any one component of this ecosystem is bound to trigger cascading consequences in the other connected components, leading to transformations in the development and deployment of AI that may not yet have been imagined.

To this end, a robust and effective national action plan for the development of AI should recognize the need for fostering and facilitating advances across all components of this rapidly shifting AI ecosystem, focusing on the fundamental concepts shaping the system without fixating on the

current circumstances and immediate expressions of emerging technologies that dominate the headlines.

## Proposed Policy Objectives

Objective 1: Promote an agile, technology-friendly, and coordinated national action plan that accelerates ambitious innovation across core areas of the AI ecosystem.

Objective 2: Develop domain-specific guidelines designed to facilitate faster adoption of AI.

Objective 3: Underscore the fundamental necessity of human trust in the AI ecosystem.

Objective 4: Ensure that every American can participate in the prosperity delivered by the AI revolution.

Objective 5: Galvanize industries, government agencies, and educational institutions into a coordinated front committed to optimizing the AI ecosystem.

## Translating Policy Objectives into Action

### 1. Promoting Integrated Technological Advances in AI

Disruptive innovations require basic scientific advancements, yet the areas of research in which such disruptions may occur are inherently unpredictable. As such, we advocate coordinated federal investments across the full scope of research areas comprising the AI ecosystem. AI policy should not pick “winners;” rather, it should promote and accelerate the conditions under which disruptive technologies emerge organically.

Two current examples readily illustrate this issue: we cannot currently anticipate when Quantum Computers will be ready for widespread use, nor can we determine whether bio-inspired computing architectures and neuromorphic computing will replace current “traditional” deep algorithms. And yet the potential impact of these technologies cannot be overstated:

- Quantum Leap. According to Scott Aaronson, “the world that allows quantum computers will be fundamentally different from a world that doesn’t.” A world that “allows” quantum computers isn’t just one with faster computers; it’s one in which formerly unsolvable problems become solvable, cryptographic foundations may need rebuilding from scratch, and scientific progress could leap forward in ways we can’t fully predict today. The ability to harness quantum phenomena changes the very rules of information processing and thus reshapes core aspects of technology, commerce, security, and scientific exploration.
- NeuroAI. Despite the rapid advance in large language models (LLMs), the foundation of intelligence is not language (only around 100,000-250,000 years’ history) but the sensorimotor knowledge acquired by the neocortex through the much longer period of evolution in mammalian brains. How to computationally model the brain’s ability to achieve goals is a fundamental challenge facing both the AI and neuroscience communities. NeuroAI, at the

intersection of these two well-established fields, is expected to be the most fruitful area for growth in science, according to Norbert Wiener.

### **Action Items**

- Identify core scientific areas and technologies that would transform AI. These include (but are not limited to) Transformative AI models, Quantum Computing, Neuromorphic Computing, (Near) Zero-energy Computing, Advanced Data Storage Architectures, High-Speed Network Communication Architectures, etc.
- Fund and promote the above core scientific areas related to AI through the appropriate Federal Agencies such as the National Science Foundation, DOE, NIST, DHS Science and Technology, and other agencies and Federal research laboratories.

## **2. Domain-Specific Policies for AI Deployment**

The plurality of the AI ecosystem requires the development of domain-specific guidelines to promote the accelerated evolution and implementation of AI. The deployment of AI in electric power grids will likely differ in expression, impact, and duration from its implementation in health care, obviating the utility of a one-size-fits-all policy approach. The development of these guidelines should encourage the investigation of policies already in place for these domains and coordinate their modification for adoption in allied domains.

### **Action Items**

- Require all Federal agencies to conduct a “technical audit” of the policies pertaining to their domain (such as power grids or health care) to identify and incorporate the potential policy changes/modifications needed to safely and effectively deploy AI ecosystem
- Considering the rapidity of the AI ecosystem evolution, the above technical audit could be mandated bi-annually (or even annually).
- Mandate AI security best practices for critical infrastructure industries (e.g., banking, health care, defense)
- Support initiatives that examine how public institutions can modernize decision-making and service delivery

## **3. Establish ‘No Harm for Americans’ Guidelines**

Trust is fundamental to the commercial success of AI innovation. By extension, negative experiences with AI-powered technologies – whether psychological or physical – will serve to erode trust in both AI and the services to which it is harnessed. Hence, we advocate policies that would aim at ensuring that no American is harmed by the rapid advances of AI.

### **Action Items**

- Incentivize U.S. government agencies to lead by example in the constructive use of AI tools and applications to improve efficiency and effectiveness, including public services and government operations. This would help to demonstrate that AI is for the People.
- Fund interdisciplinary research on governance models that enhance public trust in rapidly changing institutional and socio-technical environments
- Ensure that AI-supported analytics, decision-support and cybersecurity tools provide explainable and auditable outputs and promote iterative critical evaluation of outcomes
- Promote legislation protecting against deepfake attacks and AI-enhanced cybercrime
- Support AI encryption research to enhance privacy in cybersecurity applications

### **4. Workforce Development Policies to Promote Prosperity**

Popular anxiety regarding the disruptions that AI might precipitate in the workforce is real; so, too, is the likelihood that the rapid deployment of AI will have a substantial impact on the American labor market. To promote American prosperity and political stability, any AI policy must acknowledge this fact and respond with ambitious and innovative workforce development initiatives that match the creative power of the AI revolution itself. Institutions such as the University at Albany can play a significant role in developing and implementing such initiatives, from reskilling incumbent workers, to training graduates to be ready to participate in a world defined by AI, to developing robust curricula for K-12 students so that they can continue to shape America's AI leadership.

### **Action Items**

- Direct federal investments toward the development of AI technologies that augment and enhance the capabilities of the American workforce, providing opportunities for social mobility and wealth-building for American workers across industries and competencies
- Establish federal grants for research on labor force transitions, skill-building initiatives for incumbent workers, and employment resilience
- Increase funding for research on AI's impact on labor markets, professional mobility, and governance structures
- Develop national research initiatives that explore AI's influence on economic forecasting and public policy decision-making
- Establish ethical guidelines and accountability measures for AI-driven hiring, surveillance, criminal justice and public sector decision-making

## **5. Fostering Industry-Government-Education Institutional Cooperation**

Taken in total, the four policy objectives outlined above are best realized through the coordinated collaboration among the major players in the AI ecosystem: corporations/industry, government agencies, and educational training institutions. The federal government can catalyze the synergistic cooperation between commercial competition and deep research while fostering the development of tailored training pipelines to deliver a workforce equipped with the necessary skills and expertise to advance America's competitive edge well into the future.

### **Action Items**

- Continued investments in academic research and public education both as optimal vehicles for workforce development and as complements to industry research and development
- Safeguard American AI by making it sustainable in terms of energy requirements, prioritizing sustainable energy produced on American soil to meet the energy demands required to maintain leadership in the AI ecosystem
- Establish federal centers of excellence for interdisciplinary research on workforce trends, governance models, and social adaptation
- Leverage the federal, state, and local governments as test beds for critical public issues and government interventions by creating programs specifically focused on promoting public-private partnerships to advance the use of AI to solve public sector problems. These funding programs could be administered by agencies such as the National Science Foundation and the National Institutes of Health.
- Invest in AI-enhanced telemedicine and robotics to lower costs, improve outcomes and promote the quality of life of Americans in multiple areas, such as health care and transportation, to name a few

## **Conclusion**

The University at Albany welcomes the opportunity to collaborate in shaping national research and policy priorities that strengthen economic opportunity, workforce preparedness, and institutional resilience. Federal policies that support research in workforce transformation, institutional trust, social resilience, and technological change will ensure that national policies are informed by rigorous research and analysis. UAlbany's faculty are well-positioned to contribute to these efforts, offering insights into economic mobility, governance frameworks, and technological adaptation. Continued investment in these areas will promote national competitiveness, enhance institutional effectiveness, and ensure long-term socio-political and economic prosperity.

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